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BioHackEU26 report: Developing Open Standards for a Semantic Framework in Plant Specialized Metabolism

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BioHackathon series:
BioHackathon Europe 2026
Barcelona, Spain, 2026
Project 3

Submitted: 03 Nov 2026

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Published by BioHackrXiv.org

Introduction

Plant specialized metabolites are a major source of innovative pharmaceuticals, novel foods, and biobased natural products. However, resolving the metabolic pathways that biosynthesize these compounds remains a key scientific challenge, as plant biosynthetic genes are often genomically dispersed and duplicated, resulting in highly branched, diversified, and promiscuous underlying metabolic networks. Additionally, knowledge of plant metabolic pathways remains scattered across disconnected resources, with limited interoperable annotations, and valuable experimental multi-omics datasets remain siloed and underused. To address these gaps, we developed the Linked Open Data knowledge graph PlantMetWiki (https://plantmetwiki.bioinformatics.nl/) (Del Pup, Muller, et al., 2026), the first semantically enriched knowledge base for querying experimentally validated pathway information for plant biosynthetic reactions, including predictions of biosynthetic gene clusters. Still, researchers lack an open, standardized, and reusable AI-ready format for integrating new datasets with community knowledge in the form of a Minimum Information Standard (MIS). Additionally, there is no framework to apply such a MIS to systematically link computational evidence from bioinformatics tools and paired plant transcriptomics-metabolomics datasets to predict or validate pathway-level knowledge.

Background

Plant specialized metabolites are a major source of pharmaceuticals, food ingredients, crop protection compounds, and bio-based products (Huang & Dudareva, 2023). Discovery of plant natural products depends on resolving the enzymatic routes that produce them, yet the biosynthesis of the majority remains uncharacterized (Owen et al., 2026). The obstacle is that plant pathways resist reconstruction from any single data modality: extensive genome duplication distributes pathway genes across large, paralog-rich families that are only partially clustered, the enzymes they encode are catalytically promiscuous, and the resulting networks are highly branched and biosynthesis enzymes are shared with primary metabolism (Ji et al., 2024).

Metabolomics, transcriptomics, and genome sequencing are now generating promising data that can be integrated with dedicated multi-omics analysis pipelines to address this gap (Wolters et al., 2024). However, this evidence accumulates in domain-specific resources and individual publications that each capture one complementary facet of metabolism, such as enzymatic reactions, metabolite structures, or genomic context (Singh et al., 2022), and that are often curated to different standards and incompatible identifiers. Pathway resources such as the Plant Metabolic Network (Hawkins et al., 2025) are by design restricted to experimentally validated knowledge, and provide no mechanism for recording computational predictions, the evidence supporting them, or their confidence. Predictions are therefore published as static supplementary tables rather than as traceable assertions that can be compared across tools and revised with novel experimental evidence.

To make curated pathway knowledge reusable and machine-actionable, I developed PlantMetWiki (Del Pup, Muller, et al., 2026), a semantically enriched and queryable knowledge graph dedicated to plant biosynthetic pathways, integrating experimentally validated pathway information across hundreds of plant taxa with existing ontologies, harmonized chemical annotations, and crosslinks to biosynthetic gene clusters predicted by plantiSMASH (Del Pup, Owen, et al., 2026). PlantMetWiki provides a foundation for FAIR data representation and reuse, but there is no open, standardized format for adding new experimental datasets to this curated pool, and no framework for linking computational evidence from bioinformatics tools and from paired transcriptomics-metabolomics experiments. Extending curated knowledge with traceable predictions requires a shared representation of entities, relations, evidence types, provenance, and confidence, and an evaluative framework that the field currently lacks.

During the proposed scientific exchange, I will address this through collaboration with researchers involved in the MetaboLinkAI project at ETH Zurich, which develops FAIR and AI-ready metabolomics workflows based on knowledge graphs, federated data integration, graph machine learning, and large language models. The project will provide me with direct access to state-of-the-art developments in semantic interoperability standards, AI-assisted discovery methods, and metabolomics knowledge representation, and to (part of) the multi-omics communities whose involvement any adopted standard requires.

The work will deliver a community-supported validation schema for aligning and integrating multi-omics datasets, and an evidence-based scoring method for predictions from multi-omics pipelines, interoperable with existing standards. Together, these render plant pathway knowledge FAIR and AI-ready, representing curated pathways, omics evidence, and scored predictions in a single machine-actionable graph. In the future, our proposed workflow will be capable of supporting scalable community annotation, enabling AI-assisted discovery of plant specialized metabolic pathways through graph-based prioritization of candidate pathway components and the generation of experimentally testable biosynthetic hypotheses.

Objectives

As part of the BioHackathon Europe 2026, in this project we aim to:

1. Co-develop a community-supported Minimum Information Standard (MIS) for predicted plant pathway knowledge representation;
2. Design validation schemas to integrate experimental multi-omics datasets and predictions in alignment with the MIS;
3. Establish methods to align pathway-related predictions across bioinformatics pipelines, and approaches for scoring and prioritization of these predictions based on evidence strength, provenance, and community-defined criteria.

The project expands the plant metabolism knowledge graph PlantMetWiki with predictions from (multi-)omics plant knowledge coming from the community, which requires community-supported standards with feedback from the plant science community. We build an *evidence model* that says, for any prediction or experimental result attached to a PlantMetWiki pathway, enzyme, gene or metabolite, why it is believed (a typed evidence statement), where it comes from (tool, version, input data) and how strongly it is anchored in the knowledge graph. Users can then filter the graph by the kind of support behind a statement. The terms come from existing ontologies wherever possible; what no ontology provides is collected as candidates for new terms. These outcomes will enable scalable community annotation of pathway information, improved integration of multi-omics data, prioritization of candidate pathway components, and more reproducible and transparent bioinformatics analyses.

Implementation

All work is open and lives in the repository pathway-lod/lod-pathways-community: the specification, the discussion workflow, the validation shapes, the pilots and the generated outputs.

Questionnaire to Plant Science Labs

To ground the requirements in how the data are used, we interviewed six core laboratories that generate paired plant omics data, with a questionnaire. It asked for the key research question or goal of each laboratory, what they use to answer it (data types and analyses), which bioinformatics tools they use, and what they would like to ask a knowledge graph such as PlantMetWiki. All four questions allowed several answers. The answers are anonymised and kept in the repository together with the scripts that draw the figures (`paper/figures/`).

All six laboratories start from transcriptomics and all six want to find candidate genes; five want to reconstruct pathways, three to find enzymes and three to find or annotate metabolites (Figure 1). Orthology and comparative pathway analysis are the next most common starting points (five and four laboratories), and metabolomics is used by two. The tools reflect the same picture (Figure 2): OrthoFinder is named by five laboratories, SIRIUS, MZmine, DESeq and plantiSMASH by four each, and paired-omics tools (MEANtools, mixOmics, an in-house tool) by only two. Nine answers to the question of what to ask PlantMetWiki contained 14 questions. Most ask for genes or enzymes of a pathway or reaction step, or for comparisons of pathways and orthologs across species; others ask for routes to a compound, for genes linked to a trait or stress, for the physical clustering of genes, and for compounds (novel products, bioactivity, chlorinated products). We use these questions as test cases for the evidence model. They also show needs that the specification does not yet cover, in particular orthology, transcription factors and traits.

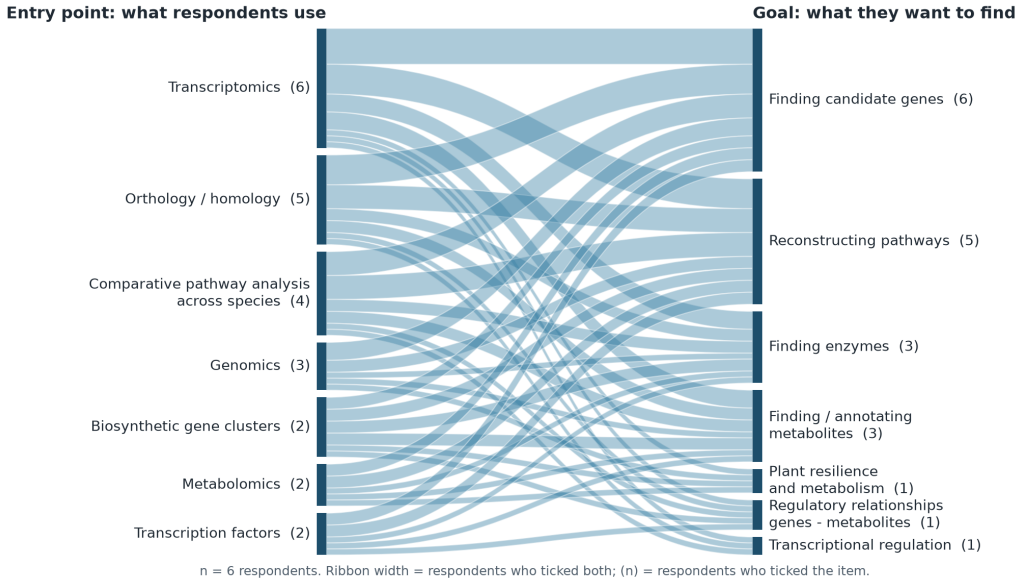


Figure 1: Questionnaire to six core laboratories: what they use (entry point) and what they want to find (goal). Ribbon width is the number of laboratories that ticked both, the number in brackets the laboratories that ticked the item.

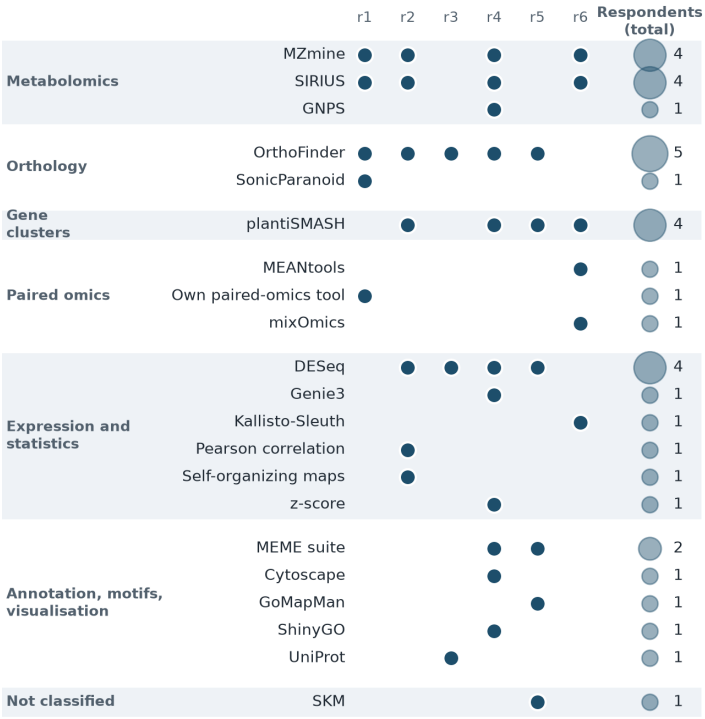
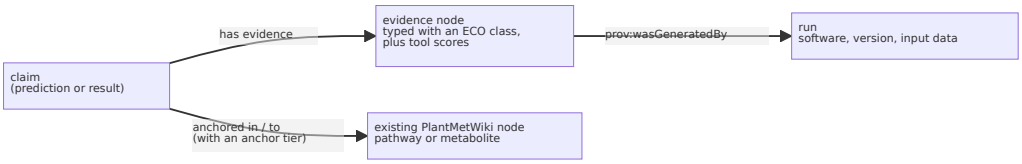


Figure 2: Bioinformatics tools named by the six laboratories, grouped by use. A dot marks a laboratory that named the tool.

The evidence model

Every prediction or experimental result is modelled in the same way, so that predictions of different kinds can be queried together. A *claim* (the prediction or result) has evidence (`sio:SIO_000772`), an evidence node typed with a class of the Evidence and Conclusion Ontology (ECO) and carrying the scores of the tool; the evidence was generated by a *run* (software, version, input data) following PROV-O; and the claim is anchored in an existing PlantMetWiki node (pathway or metabolite) with an *anchor tier* that records how strongly it is tied to the knowledge graph (for example, a gene-level link is stronger than a link only through an EC number).



Use cases to support

The use cases are the kinds of predictions and data that we want PlantMetWiki to be able to hold. Each has an acronym that is also the prefix of its requirement IDs in the specification (for example `PO-03` is the third requirement of the paired omics use case).

Table 1: Use cases supported by the framework, with the pilot dataset used to test each. Cases 5 and 6 are emphasised in our working list.

#	Use case	Specification	Test data	Pilot
1	New pathway from the literature	not yet a use case of its own	Ipecac alkaloid biosynthesis in two evolutionarily distant plants (Colinas et al.)	`pilots/ipecac_overlap`
2	PlantCyc (PMN) inferred enzyme function	PMN	PlantCyc evidence codes on PlantMetWiki nodes	`shape_constraints/rdf/plantcyc.ttl`
3	plantiSMASH-predicted biosynthetic gene clusters (BGCs)	BGC	plantiSMASH 2.0 precalculated database, about 30,400 BGCs in 430 genomes	`pilots/plantismash_overlap`
4	Transcriptomics	TX	covered inside the paired data of cases 1 and 6	no separate pilot
5	**Metabolomics**	MET	Brassicaceae metabolic fingerprinting (Wolters et al. 2026)	`pilots/metabolomics_overlap`
6	**Paired omics** (transcript-metabolite associations)	PO	Tomato data of the MEANtools paper (Singh et al., 2025)	`pilots/meantools_overlap`
7	"Unsupervised" graph neural network (GNN) predictions of conversion steps	UN	none yet	none

Working groups

The work is organised in four groups, which correspond to the sections of the Results:

1. **Terms.** Discuss and curate the proposed terms and the gaps in the specification, and identify the terms to propose to an existing ontology.
2. **Pilots.** Test the alignment of the use case datasets with PlantMetWiki.
3. **Visualization and extraction of results.** Show what the aligned data say, and rank or score predictions.
4. **Submission system (optional).** Build and test an automated way to submit, integrate and approve new predictions, with decentralised governance.

The pipeline

The work follows six steps, each with its own folder and documentation in the repository.

Table 2: The six steps of the pipeline.

Step	What happens	Where
1. Use cases	Choose the kinds of predictions and experimental data to align with PlantMetWiki, and what each needs to say	`specification/`
2. Specification	For each use case: the requirements (what must be said), the candidate ontology terms and the gaps. One row per requirement and term in a table (`ontology_terms.csv`, the source of truth), the text in `SPECIFICATION.md`. Term IDs and labels are checked against the EBI Ontology Lookup Service (OLS)	`specification/`
3. Community discussion	The table is shown as a web dashboard; discussion happens in GitHub issues; maintainers' labels turn decisions into pull requests (see below)	`.github/`, dashboard
4. Shape constraints	The chosen terms become SHACL shapes that say what a statement must contain before it is added to the knowledge graph. Problems are either rejected (violations) or flagged (warnings)	`shape_constraints/`
5. Pilots	Each use case is tried on a real dataset	`pilots/`
6. Outputs	RDF (Turtle) validated against the shapes, a sample graph, a local placeholder vocabulary (`lodp:`) for terms that no ontology provides, and the terms to propose to larger ontologies	`shape_constraints/rdf/`, `examples/`

The specification dashboard: how terms are discussed and approved

The specification is shown as an interactive dashboard at https://pathway-lod.github.io/lod-pathways-community/. It lists every requirement with its status, decision, owner and candidate terms, with filters by use case, decision, owner, action and ontology, and charts of the number of terms per ontology and of the coverage. Anyone can take part:

1. **Find a requirement** with the search box or the filters, and select its ID to see the candidate terms.
2. **Start a discussion** with the buttons on the requirement. They open a GitHub issue form

that is already filled in: *ask a question*, *suggest a term*, or *propose a new requirement*. Issues carry the label `ontology-terms` and the requirement ID in the title (for example `[PMN-03]`).

3. **Take a requirement on** by ticking the volunteering box of the form, or ask a maintainer to assign the issue.
4. **Maintainers decide with labels.** `decision:agreed`, `decision:rejected`, `decision:proposed` or `decision:open` on an issue makes a GitHub Action open a pull request that changes the `decision` of that requirement in the table. The label `accept-term` on a term suggestion adds the term as a new row (its label and IRI are looked up in OLS) and `decision:rejected` marks only that suggested term as rejected.
5. **Review and merge.** Nothing reaches the main branch without a reviewed pull request. Only people with triage or write access can set labels, so only they can trigger a change. The issue text is read as data and is checked against the lists of the specification.
6. **The dashboard is regenerated** from the table when a pull request is merged, and the checks on every pull request keep the generated files and the table in sync.

Terms that no ontology provides are rows of the same table, with the action `local_term` while a local placeholder is used and `request_term` when a term should be proposed to an existing ontology such as ECO, STATO, PSI-MS or EDAM.

How the candidate terms were found and checked

For each requirement we searched candidate terms in the Ontology Lookup Service (OLS4) through its search API (`https://www.ebi.ac.uk/ols4/api/search`). The query was a free-text phrase taken from the requirement (for example "co-expression", "mass error" or "area under the ROC curve"), usually restricted to one ontology (ECO, PSI-MS, EDAM, SIO, STATO, SO, CHEMINF, CHMO, SWO, AIO, OBI and others), and we read the first four to six hits (identifier, label and sometimes the definition). The searches were run interactively from the command line with the help of an AI coding assistant (Claude Code), and the scripts were not kept. We reconstructed the phrases afterwards from the log of the working session: 223 distinct phrases in 12 rounds (Appendix B). They are the starting point and do not have to match the final terms, because terms were also taken from other sources (the BioCyc evidence ontology for the PlantCyc evidence codes, the papers and supplementary tables of the pilot data sets) and many phrases led to no term or to a term that was not used. The hits were chosen by reading labels and definitions, and the role (primary, alternative, stand-in, local, gap) and coverage (exact, partial, none) of each term were set by hand. The table was then reviewed manually by the project members and is revised by the community through the dashboard (issues, labels and reviewed pull requests, as described above).

The script `check_terms.py` of the repository keeps the term table consistent and generates everything derived from it. It uses only the Python standard library (`requests` for the online check). It performs four tasks:

1. **Validation** of the term table without network access: unique row identifiers; requirement identifiers that match their use case (for example `PMN-03`); allowed values for element type, role, coverage, decision and action; a valid CURIE whose prefix belongs to the named ontology; and the rules between columns (a gap has no term and coverage `none`; a local term belongs to the local vocabulary and has coverage `none`; primary and alternative terms have coverage `exact` or `partial`; a term appears once per requirement; `value_of` points to a term that is in the table).
2. **Online check** (`--online`): every term is looked up in OLS by an exact search on its identifier. A term that is not found or is obsolete is an error; a stored label that differs from the one in OLS is a warning. Vocabularies that are not in OLS (PROV-O, SKOS, Dublin Core Terms, RDF, WikiPathways, identifiers.org and our local placeholder vocabulary) are not looked up.
3. **Update** (`--update`): writes the label, IRI and date of the check of new or changed

terms into the table, so that these columns come from OLS and not from typing.
4. **Rendering** (`--render`): regenerates the tables of the specification document and the dashboard from the table, and `--check-render` fails when a generated file is out of date.

GitHub Actions run these steps for every pull request and for every term suggestion that a maintainer accepts with a label, and regenerate the dashboard before it is published, so that the published specification always follows the term table.

Results

1. Discussion and curation of proposed and gap terms

At the time of writing the specification has seven use cases and 119 requirements, described by 188 rows of candidate terms (TODO: update the numbers before submission; they are generated from `ontology_terms.csv`). Of these rows, 75 are a primary term, 37 an alternative, 18 a stand-in, 49 a local term and 9 mark a gap where no term exists. Most requirements still wait for a decision (104 rows proposed, 82 open, 2 rejected). The dashboard makes these numbers current, and is the place where co-authors and the wider community discuss and approve the terms.

Terms to propose to an existing ontology. Seven requirements are marked `request_term`: for these we have searched OLS and found no suitable term, and we intend to propose one to an existing ontology (for example EDAM, ECO, STATO or PSI-MS) instead of keeping a local term. An example is metabolite-transcript correlation evidence, for which `ECO:0007590` is only the closest term. TODO: list the final terms to propose, the ontology and the status of each request.

2. Test alignment of use case datasets (pilots)

Each pilot checks how much of a real dataset can be aligned with PlantMetWiki, and what evidence is available. TODO: add the headline numbers of the first three pilots from their READMEs.

- **Metabolomics** (`pilots/metabolomics_overlap`): LC-MS/MS annotations of 14 Brassicaceae species (Wolters et al. 2026) matched to PlantMetWiki metabolites on the first InChIKey block, with six evidence tiers from the GNPS library match to MS2Query analogues; the three annotation tools are combined in a consensus (agreement of three tools counts 1.0, two tools 0.5).
- **Paired omics** (`pilots/meantools_overlap`): gene-metabolite associations of MEANtools (Singh et al., 2025) functional clusters in tomato, with genes mapped to PlantMetWiki pathways in three tiers (locus, UniProt, EC) and metabolite features by exact mass (MS1 only), compared against a chance background.
- **BGC predictions** (`pilots/plantismash_overlap`): plantiSMASH 2.0 (Del Pup, Owen, et al., 2026) clusters in 430 genomes linked to PlantMetWiki genes and pathways in three tiers.
- **New pathway from the literature** (`pilots/ipecac_overlap`): the paired ipecac alkaloid data (Colinas et al.). Of the 12 metabolites of the paper, 6 are in PlantMetWiki (secologanin exactly; three with a protonation difference only; two on the connectivity level only) and 6 are absent, among them all those of the acid route. The pathway of emetine biosynthesis (PWY-7135) has 8 conversions, of which 5 are linked to a UniProt enzyme, one only to a PlantCyc monomer, and two to no enzyme. The cloned *Carapichea* genes are 94-98% identical to enzymes already in this pathway, the *Alangium* genes at most 46%. The heterologous expression and infiltration experiments give reaction evidence for the step without an enzyme and for new acid-route compounds

(75 comparisons, 41 with a higher product level). The study is targeted metabolomics of known compounds, so it tests reaction-level evidence and not spectral matching.

Pilots on the other use cases (PMN, TX, UN) are still WIP.

3. Ranking and Scoring of Predictions

The pilots already rank evidence in tiers (for example `gene_direct` over `gene_uniprot` over `gene_ec`, and the six tiers of the metabolite annotations) and the consensus of three annotation tools gives a score between 0 and 1. A scoring framework that ranks candidate pathway components by evidence strength, completeness, provenance and reproducibility, and its benchmark against the experimentally validated pathways of PlantMetWiki, is still to be defined (TODO).

4. Submission system (optional)

The discussion workflow of the specification (issue forms, labels and Actions that open pull requests) is already an automated, reviewed way to approve *terms*. TODO: test an automated system to submit, validate (against the SHACL shapes) and approve new *predictions*; a design note and a local test of Nanodash, a tool for publishing signed nanopublications, are being explored for this.

Governance. We aim to improve decentralised governance: who may propose, who may approve, and how decisions are traced. Today only people with triage or write access to the repository can approve, and every change passes a reviewed pull request. TODO: describe the governance model agreed by the community.

Discussion

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Acknowledgements

We thank Nicola Zamboni and Adriano Rutz for the supervision of Elena Del Pup during the preparation of the BioHackathon and the writing of this report. ...

References

Appendix B: Phrases searched in the Ontology Lookup Service

The phrases used to start the search for candidate terms (see "How the candidate terms were found and checked"), reconstructed from the log of the working session. The search was restricted to one ontology where one is named, otherwise it covered all ontologies of OLS. The final terms were selected and reviewed manually, so they do not necessarily appear here. The same table is in the repository (`specification/ols_search_phrases.csv`).

Table 3: Phrases searched in the Ontology Lookup Service, in the order of the search rounds.

Round	Topic	Search phrase	Ontology
1	evidence types, gene clusters, similarity	biosynthetic gene cluster	so
1	evidence types, gene clusters, similarity	gene cluster	so

Round	Topic	Search phrase	Ontology
1	evidence types, gene clusters, similarity	computational evidence used in automatic assertion	eco
1	evidence types, gene clusters, similarity	match to sequence model evidence	eco
1	evidence types, gene clusters, similarity	sequence similarity evidence	eco
1	evidence types, gene clusters, similarity	genomic context evidence	eco
1	evidence types, gene clusters, similarity	correlation evidence	eco
1	evidence types, gene clusters, similarity	expression pattern evidence	eco
1	evidence types, gene clusters, similarity	co-expression	eco
1	evidence types, gene clusters, similarity	mass spectrometry evidence	eco
1	evidence types, gene clusters, similarity	combinatorial computational and experimental evidence	eco
1	evidence types, gene clusters, similarity	author inference	eco
1	evidence types, gene clusters, similarity	curator inference	eco
1	evidence types, gene clusters, similarity	imported information	eco
1	evidence types, gene clusters, similarity	biosynthetic gene cluster	mibig
1	evidence types, gene clusters, similarity	percent identity	sio
1	evidence types, gene clusters, similarity	overlap	sio
2	evidence, paired omics, annotation tools	coexpression evidence	eco
2	evidence, paired omics, annotation tools	statistical association evidence	eco
2	evidence, paired omics, annotation tools	gene neighbors evidence used in automatic assertion	eco
2	evidence, paired omics, annotation tools	machine learning	eco
2	evidence, paired omics, annotation tools	inference from background scientific knowledge	eco
2	evidence, paired omics, annotation tools	genomic context evidence used in automatic assertion	eco

Round	Topic	Search phrase	Ontology
2	evidence, paired omics, annotation tools	Pearson correlation coefficient	stato
2	evidence, paired omics, annotation tools	Spearman	stato
2	evidence, paired omics, annotation tools	mutual rank	stato
2	evidence, paired omics, annotation tools	p-value	obi
2	evidence, paired omics, annotation tools	has version	swo
2	evidence, paired omics, annotation tools	software version	iao
2	evidence, paired omics, annotation tools	antiSMASH	edam
2	evidence, paired omics, annotation tools	gene cluster prediction	edam
2	evidence, paired omics, annotation tools	metabolic pathway prediction	edam
2	evidence, paired omics, annotation tools	tandem mass spectrometry	msio
3	paired omics	coexpression	eco
3	paired omics	gene expression correlation	eco
3	paired omics	correlation	eco
3	paired omics	statistical inference	eco
3	paired omics	network	eco
3	paired omics	clustering	eco
3	paired omics	mutual rank	all
3	paired omics	co-expression network	all
3	paired omics	gene-metabolite association	all
3	paired omics	correlation coefficient	stato
3	paired omics	association	sio
3	paired omics	cluster	edam
3	paired omics	gene-chemical association	sio
3	paired omics	chemical-gene association	sio
3	paired omics	has evidence	sio
3	paired omics	has value	sio
3	paired omics	refers to	sio
3	paired omics	p-value	sio

Round	Topic	Search phrase	Ontology
3	paired omics	gene	sio
3	paired omics	chemical entity	sio
3	paired omics	expression pattern evidence used in automatic assertion	eco
3	paired omics	clustering	edam
3	paired omics	gene expression correlation analysis	edam
4	gene clusters, evidence	gene cluster	edam
4	gene clusters, evidence	biosynthetic gene cluster	edam
4	gene clusters, evidence	secondary metabolite	edam
4	gene clusters, evidence	combinatorial computational evidence used in automatic assertion	eco
4	gene clusters, evidence	imported information used in automatic assertion	eco
4	gene clusters, evidence	sequence similarity evidence used in automatic assertion	eco
4	gene clusters, evidence	match to sequence model evidence used in automatic assertion	eco
4	gene clusters, evidence	has part	ro
4	gene clusters, evidence	overlaps	ro
4	gene clusters, evidence	percent identity	all
4	gene clusters, evidence	similarity score	sio
5	PlantCyc evidence codes	EV-EXP-IDA	eco
5	PlantCyc evidence codes	EV-EXP-IEP	eco
5	PlantCyc evidence codes	EV-COMP-HINF-SEQ-ORTHOLOGY	eco
5	PlantCyc evidence codes	EV-IC	eco
5	PlantCyc evidence codes	EV-AS-TAS	eco
5	PlantCyc evidence codes	EV-COMP-AINF	eco
5	PlantCyc evidence codes	direct assay evidence used in manual assertion	eco

Round	Topic	Search phrase	Ontology
5	PlantCyc evidence codes	expression pattern evidence used in manual assertion	eco
5	PlantCyc evidence codes	sequence orthology evidence used in manual assertion	eco
5	PlantCyc evidence codes	traceable author statement used in manual assertion	eco
5	PlantCyc evidence codes	inferred by curator	eco
5	PlantCyc evidence codes	automatic assertion computational inference	eco
6	evidence, identification confidence	experimental evidence	eco
6	evidence, identification confidence	metabolite identification confidence level	all
6	evidence, identification confidence	metabolomics standards initiative level	all
6	evidence, identification confidence	machine learning method evidence used in automatic assertion	eco
6	evidence, identification confidence	spectral library match evidence	eco
6	evidence, identification confidence	deep learning neural network method evidence used in automatic assertion	eco
6	evidence, identification confidence	compound identification confidence level	ms
7	metabolomics, evidence model	spectral library search	all
7	metabolomics, evidence model	spectral library search	edam
7	metabolomics, evidence model	mass spectra similarity	edam
7	metabolomics, evidence model	positive scan	ms
7	metabolomics, evidence model	negative scan	ms
7	metabolomics, evidence model	retention time	ms
7	metabolomics, evidence model	selected ion m/z	ms
7	metabolomics, evidence model	adduct ion	ms

Round	Topic	Search phrase	Ontology
7	metabolomics, evidence model	InChIKey	cheminf
7	metabolomics, evidence model	topic model	all
7	metabolomics, evidence model	topic modelling	edam
7	metabolomics, evidence model	substructure	cheminf
7	metabolomics, evidence model	mass spectrometry feature	all
7	metabolomics, evidence model	SEPIO	all
7	metabolomics, evidence model	evidence line	sepio
7	metabolomics, evidence model	assertion	sepio
7	metabolomics, evidence model	probability value	sio
7	metabolomics, evidence model	software	swo
7	metabolomics, evidence model	evidence	eco
7	metabolomics, evidence model	nanopublication	all
7	metabolomics, evidence model	molecular network	all
7	metabolomics, evidence model	fragmentation pattern	all
8	evidence (manual assertion)	computational evidence used in manual assertion	eco
8	evidence (manual assertion)	evidence used in manual assertion	eco
8	evidence (manual assertion)	computational inference used in manual assertion	eco
8	evidence (manual assertion)	author statement used in manual assertion	eco
8	evidence (spectral, unsupervised)	Mass2Motif	all
8	evidence (spectral, unsupervised)	MS2LDA	all
8	evidence (spectral, unsupervised)	unsupervised	eco
8	evidence (spectral, unsupervised)	unsupervised	all
8	evidence (spectral, unsupervised)	clustering evidence	eco
8	evidence (spectral, unsupervised)	spectral library	eco
8	evidence (spectral, unsupervised)	mass spectral library	eco

Round	Topic	Search phrase	Ontology
8	evidence (spectral, unsupervised)	spectrum matching	eco
8	evidence (spectral, unsupervised)	molecular family	all
8	evidence (spectral, unsupervised)	molecular networking	all
8	evidence (spectral, unsupervised)	mass spectrometry	eco
9	machine learning predictions	graph neural network	aio
9	machine learning predictions	link prediction	all
9	machine learning predictions	knowledge graph embedding	all
9	machine learning predictions	hyperparameter	all
9	machine learning predictions	machine learning model	aio
9	machine learning predictions	model evaluation	all
9	machine learning predictions	area under the ROC curve	all
9	machine learning predictions	area under curve	stato
9	machine learning predictions	average precision	all
9	machine learning predictions	precision at k	all
9	machine learning predictions	training data set	all
9	machine learning predictions	data split	aio
9	machine learning predictions	cross validation	stato
9	machine learning predictions	prediction score	all
9	machine learning predictions	protein language model	all
9	machine learning predictions	predicted	sio
9	machine learning predictions	confidence score	all
9	machine learning predictions	model	iao
9	machine learning predictions	data transformation	obi
10	transcriptomics	z-score	all
10	transcriptomics	z-score	stato
10	transcriptomics	standard score	all
10	transcriptomics	DESeq2	all
10	transcriptomics	differential expression analysis	edam

Round	Topic	Search phrase	Ontology
10	transcriptomics	differential gene expression	all
10	transcriptomics	log fold change	all
10	transcriptomics	fold change	all
10	transcriptomics	false discovery rate	stato
10	transcriptomics	Benjamini-Hochberg	all
10	transcriptomics	adjusted p-value	all
10	transcriptomics	Wald test	all
10	transcriptomics	RNA sequencing assay	obi
10	transcriptomics	RNA-seq	all
10	transcriptomics	read count	all
10	transcriptomics	transcripts per million	all
10	transcriptomics	RPKM	all
10	transcriptomics	FPKM	all
10	transcriptomics	normalized read count	all
10	transcriptomics	gene expression level	all
10	transcriptomics	expression profile	all
10	transcriptomics	RNA-sequencing evidence	eco
10	transcriptomics	differential expression evidence	eco
10	transcriptomics	co-expression	all
10	transcriptomics	coexpression network	all
10	transcriptomics	rank correlation	all
10	transcriptomics	variance stabilizing transformation	all
10	transcriptomics	library size	all
10	transcriptomics	replicate	all
10	transcriptomics	biological replicate	obi
10	transcriptomics	treatment	all
10	transcriptomics	plant structure	po
10	transcriptomics	leaf	po
10	transcriptomics	experimental factor	efo
10	transcriptomics	base mean	all
10	transcriptomics	dispersion	stato
10	transcriptomics	standard error	stato
10	transcriptomics	mean	stato
10	transcriptomics	log2	all
11	metabolomics acquisition and processing	Orbitrap Exploris 120	ms
11	metabolomics acquisition and processing	instrument model	ms
11	metabolomics acquisition and processing	heated electrospray ionization	ms
11	metabolomics acquisition and processing	electrospray ionization	ms

Round	Topic	Search phrase	Ontology
11	metabolomics acquisition and processing	normalized collision energy	ms
11	metabolomics acquisition and processing	higher energy beam-type collision-induced dissociation	ms
11	metabolomics acquisition and processing	resolution	ms
11	metabolomics acquisition and processing	scan window	ms
11	metabolomics acquisition and processing	data-dependent acquisition	ms
11	metabolomics acquisition and processing	dynamic exclusion	ms
11	metabolomics acquisition and processing	exclusion list	ms
11	metabolomics acquisition and processing	isolation window	ms
11	metabolomics acquisition and processing	MZmine	ms
11	metabolomics acquisition and processing	SIRIUS	ms
11	metabolomics acquisition and processing	feature detection	ms
11	metabolomics acquisition and processing	peak picking	ms
11	metabolomics acquisition and processing	alignment	ms
11	metabolomics acquisition and processing	blank subtraction	ms
11	metabolomics acquisition and processing	feature-based molecular networking	edam
11	metabolomics acquisition and processing	molecular networking	edam
11	metabolomics acquisition and processing	mzmine	swo

Round	Topic	Search phrase	Ontology
11	metabolomics acquisition and processing	MS2LDA	swo
11	metabolomics acquisition and processing	liquid chromatography	chmo
11	metabolomics acquisition and processing	untargeted metabolomics	chmo
11	metabolomics acquisition and processing	peak area	all
11	metabolomics acquisition and processing	feature abundance	all
11	metabolomics acquisition and processing	ion intensity	all
11	metabolomics acquisition and processing	ClassyFire	all
11	metabolomics acquisition and processing	NPClassifier	all
11	metabolomics acquisition and processing	chemical taxonomy	all
11	metabolomics acquisition and processing	precursor exclusion	all
11	metabolomics acquisition and processing	mass spectral motif	all
11	metabolomics acquisition and processing	substructure motif	all
11	metabolomics acquisition and processing	heated electrospray	all
11	metabolomics acquisition and processing	blank	all
12	mass-only annotation	liquid chromatography mass spectrometry evidence	eco
12	mass-only annotation	accurate mass	ms
12	mass-only annotation	exact mass	chmo
12	mass-only annotation	mass-to-charge ratio	ms
12	mass-only annotation	MS1 spectrum	ms
12	mass-only annotation	full scan	ms
12	mass-only annotation	mass error	ms

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Appendices

Appendix A: Original project plan (from the proposal)

The plan consists of three interconnected work packages corresponding to the project objectives. The BioHackathon Europe 2026 (9 - 14 November 2026, Barcelona) extends this work to a broader international community.

Community requirements and draft semantic framework (Objective 1)

The first month establishes what a minimum information specification for predicted plant biosynthetic pathway knowledge must contain. Requirements will be collected through a structured questionnaire, disseminated to an identified list of stakeholders across European institutions, and through targeted interviews with researchers in metabolomics, plant biology, semantic web technologies, and the relevant ELIXIR communities. Questions will cover data modalities in routine use, metadata that is recorded in practice, analytical workflows, predictive tools, and datasets available for reuse.

In parallel, existing ontologies, controlled vocabularies, and interoperability resources will be reviewed, including WikiPathways (Zdouc et al., 2025), BridgeDb (Rutz et al., 2026), Identifiers.org, and the Ontology Lookup Service. A gap analysis will identify concepts that no current vocabulary expresses to describe evidence, provenance, and confidence, and will document where existing terms can be reused. The specification will follow the structure of MIBiG (Singh et al., 2025) and MITE (Sansone et al., 2019), which are already familiar to the natural product community, while being built directly on a machine-actionable RDF-based model.

In parallel, I will participate in the meetings of the MetaboLinkAI project and present PlantMetWiki (Del Pup, Muller, et al., 2026) and the project objectives to partners to identify synergies with existing tools and involve them in the standard development process. Together with the host laboratory, I will select one paired transcriptomics-metabolomics dataset as the proof-of-concept for Months 2 and 3, prioritizing a species for which PlantMetWiki already holds curated pathways, so that predictions can later be assessed against known biology.

Expected outputs: stakeholder requirements report; gap analysis of existing vocabularies; draft minimum information specification expressed as an RDF data model; selected proof-of-concept dataset.

Prototype validation schema (Objective 2)

The second month implements and tests the specification to prototype an end-to-end workflow connecting pathway knowledge, metabolomics data, transcriptomics evidence, and computational predictions within a common semantic framework. Semantic data shapes (SHACL or ShEx) will be written to encode the required metadata, provenance, quality criteria, and interoperability constraints, giving machine-actionable validation of multi-omics datasets (transcriptomics and metabolomics).

The selected transcriptomics-metabolomics dataset will be transformed into a semantically enriched representation and linked to pathway entities in PlantMetWiki following Linked Open Data principles and the FAIR guiding principles (Agrawal et al., 2024). Achieving this requires resolving identifier mismatches, provenance modeling, and vocabulary alignment between resources, which will be supported by the expertise of the Zamboni Lab and MetaboLinkAI partners.

Interactions with MetaboLinkAI researchers and collaborators from the ELIXIR BioHackathon project "Developing Open Standards for a Semantic Framework in Plant Specialized Metabolism" (9 - 14 November 2026 in Barcelona) will keep the schema aligned with parallel developments

in federated querying, semantic enrichment, and FAIRification workflows, and will ensure that what is prototyped here is what the community will later be asked to adopt.

Expected outputs: validation schema implementing the specification; tested end-to-end FAIR integration workflow; semantically enriched proof-of-concept dataset; demonstration of interoperable querying across PlantMetWiki and multi-omics-supported predictions.

Evidence integration, benchmarking, and dissemination (Objective 3)

The final month focuses on integrating computational predictions and experimental evidence into a unified framework for pathway prioritization and hypothesis generation. Predictions for the proof-of-concept species by plantiSMASH (Del Pup, Owen, et al., 2026) and MEANtools (van Iersel et al., 2010) will be mapped to a common semantic representation and aligned with experimental observations.

A scoring framework will be defined to rank candidate pathway components by evidence strength, completeness, provenance, and reproducibility. Its performance will be benchmarked using the experimentally validated pathways already curated in PlantMetWiki as ground truth. This provides an objective measure of whether evidence-based scoring adds discriminative power, and yields a prioritised list of candidate components that are well supported but not yet experimentally described, which are of direct interest for follow-up work in both laboratories.

The resulting workflows, standards, and validation approaches will be reviewed with collaborators from MetaboLinkAI, ELIXIR, and WikiPathways, released openly and registered on FAIRsharing (Wilkinson et al., 2016) to maximize visibility, reuse, and long-term sustainability. The ELIXIR BioHackathon in Barcelona (9-14 November 2026) will extend the outreach to a broader international community to refine, test, and adopt them, beyond the duration of the fellowship.

Expected outputs: benchmarked scoring scheme; prioritized candidate pathway components for the proof-of-concept species; openly released specification, schemas, and workflows.

The three packages are sequential but independently deliverable: the specification and gap analysis stand alone as a community contribution even if integration proves more complex than anticipated, and the validation schema is useful independently of the scoring results. Overall, the work will contribute towards establishing a FAIR and interoperable framework for integrating experimentally validated pathway knowledge, multi-omics evidence, and computational predictions, enabling more reproducible and scalable discovery of plant specialized metabolic pathways.